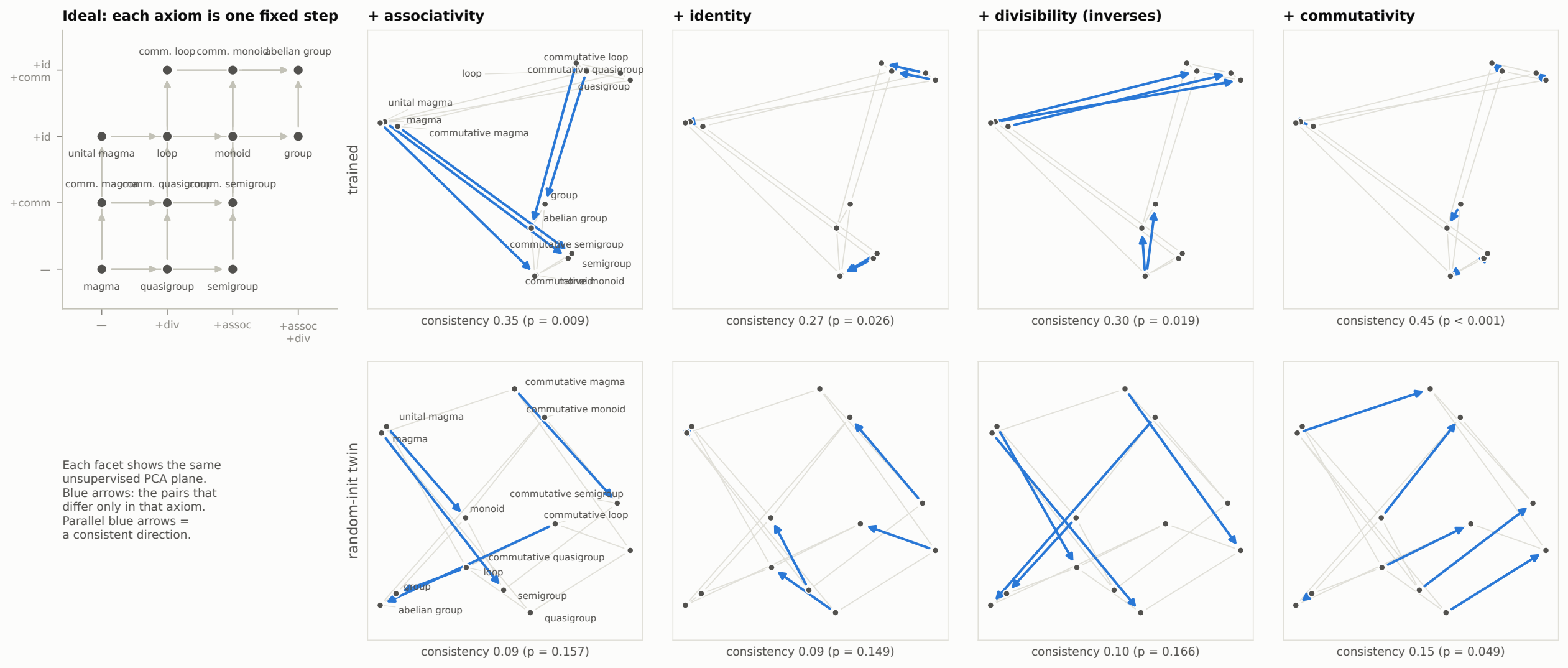
Ideal: each axiom is one fixed step
+ associativity
+ identity
+ divisibility (inverses)
+ commutativity
comm. loop
comm. monoid
abelian group
+id +comm
+id
unital magma
loop
monoid
group
comm. magma
comm. quasigroup
comm. semigroup
+comm
magma
quasigroup
semigroup
—
+div
+assoc
+assoc +div
trained
commutative loop
commutative quasigroup
loop
quasigroup
unital magma
magma
commutative magma
group
abelian group
commutative semigroup
semigroup
commutative monoid
consistency 0.35 (p = 0.009)
consistency 0.27 (p = 0.026)
consistency 0.30 (p = 0.019)
consistency 0.45 (p < 0.001)
random-init twin
commutative magma
commutative monoid
unital magma
magma
commutative semigroup
monoid
commutative loop
commutative quasigroup
loop
group
semigroup
abelian group
quasigroup
consistency 0.09 (p = 0.157)
consistency 0.09 (p = 0.149)
consistency 0.10 (p = 0.166)
consistency 0.15 (p = 0.049)
Each facet shows the same unsupervised PCA plane. Blue arrows: the pairs that differ only in that axiom. Parallel blue arrows = a consistent direction.